

LEONARD AARONS-DITSON

RESEARCH ENGINEERING | AGENT EVALS | TOOL USE | SCIENTIFIC REASONING

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PROFILE

Physics-trained researcher and engineer interested in how agentic systems behave when tools, evaluators and execution environments become part of the experiment. **I own the research question, protocol, acceptance criteria and interpretation; coding agents accelerate implementation.** Recent work asks when models know to use tools, whether agent cross-review catches real defects, and how an evaluation can produce a convincing but misleading result. UCL quantum-information background and startup data-science experience. Interested in evals/post-training work where distinguishing capability gain from harness artefact is the hard part.

SELECTED RESEARCH & AGENT SYSTEMS

[Agentic Physics Bench](#) | *Controlled Tool-Use Evaluation in Scientific Reasoning* | Python, Claude Code

Research question: *When a frontier model has access to external computation, does it recognise when the tool is actually useful?*

- Designed a frozen, hash-verified no-tool / optional-tool / required-tool study: 162 scored episodes (324 calls), deterministic grading and no retries. A trace audit exposed hidden advisor contamination in 31/48 development calls, so I added per-call controls before scoring.
- Correctness hit the ceiling (54/54 in every condition). I therefore reported behaviour rather than manufacturing a performance claim: optional tool requests shifted from 0/12 in V1 to 54/54 in V2, with the cause explicitly left unresolved. The main lesson was that eval integrity was harder than the benchmark itself.

[Agent Workflow Orchestrator](#) | *Cross-Review Between Autonomous Coding Agents* | Python, Git

Research question: *Can one coding agent catch meaningful defects that another agent and its acceptance tests miss?*

- Designed a Codex/Claude workflow using isolated Git worktrees, frozen diffs, cross-review, revision and deterministic gates; 150 offline tests. In live runs, Codex identified a Unicode lowercasing defect in Claude code that had passed all four acceptance tests, and Claude corrected it.
- The review layer also failed in useful ways: unusable reviews and an execution artefact caused candidates to be rejected. I treated those as evidence about the reliability of agent review, not noise to hide.

[Institutional Workbench](#) | *Role-Prompting and Protocol-Reliability Study* | Python, Pydantic

Research question: *Do role prompts produce repeatable behavioural differences, or do protocol failures dominate the result?*

- Ran a 96-call pilot across tasks, roles and two CLIs. Protocol reliability dominated the apparent differences (Codex 48/48 schema-valid; Claude Sonnet 15/48), so I reported a protocol problem rather than a model ranking or role benefit.
- A preregistered 1,000-call follow-up failed calibration 60/60 because the live CLI rejected a configuration path that mocks had never exercised. Under the pre-specified no-retry rule, I stopped rather than repair the study after seeing the outcome.

Additional Agent Engineering

- Grounded CV-Planning Pipeline:** Codex structured-output planner on 257 real job listings; deterministic fact checks rejected the first plan in 151/251 evaluable runs when it introduced unsupported claims, forcing re-planning.
- Agent Context Router:** bounded, provenance-cited context packets with compare-and-swap writes; published a negative synthetic result when a label-free route picker underperformed BM25 top-1 (21/31 vs 24/31).

EXPERIENCE

Data Scientist - Sticky (Startup), London | Nov 2020 - Dec 2021 | Python/ML on messy customer, payment and product data; investigated SaaS and operational problems and explained findings to non-technical staff.

Additional experience: 999 Emergency Call Handler, East of England Ambulance Service NHS Trust (Jan-Jun 2026); Medical Laboratory Assistant, Imperial College Healthcare NHS Trust (Sep 2023-Nov 2024).

EDUCATION & RESEARCH

University College London | Postgraduate Certificate in Physics, Distinction (2021) | BSc Physics (2018)

Advanced Quantum Theory 93.5%. Postgraduate quantum-information research activity in spin-chain communication and open-system density-matrix dynamics; UCL talk on quantum simulators.

TECHNICAL

Agents & Evaluation: experimental/eval design, tool-use evaluation, trace analysis, deterministic graders, structured outputs, prompt/protocol design, Codex CLI, Claude Code, JSON Schema, Pydantic. **Engineering & Data:** Python, pytest, Git, GitHub Actions, pandas, NumPy, scikit-learn, DuckDB, Streamlit.

Career context: Cancer disrupted my route; I continued research and work while ill and returned to full-time NHS work. Immediately available.